

Probabilistic Intelligence & Deterministic Governance

Why Mental Health and Companion AI Demand Transport-Layer Controls

An Architectural Guideline for Support, Wellness, and Clinical-Adjacent AI

SASKI Institute PBC | Stephen Calhoun | September 4, 2026

Executive Summary

For the past three years, mental health applications, wellness chatbots, and companion AI platforms have attempted to solve safety and compliance inside the language model. System prompts instructing bots to "act like a supportive listener but never give medical advice," reinforcement learning from human feedback (RLHF), and secondary safety classifiers have all been deployed to make probabilistic models police themselves.

A growing body of empirical research shows that model-internal safeguards alone cannot provide the reproducible enforcement guarantees required for safety-critical clinical and psychosocial support.

Language models operate over continuous vector spaces where safety boundaries are smooth gradients rather than discrete security perimeters. As a result, in-model safety mechanisms fail silently, unpredictably, and in ways that cannot be audited after the fact. In a casual consumer application, a safety failure is an inconvenient bug; in mental health and companion software, where interactions intersect with acute distress, suicidal ideation, and strict state laws, it creates direct legal liability and uninsurable risk.

Mental health AI does not have a model-alignment problem; it has an infrastructure problem. Statutory frameworks such as California SB 243, New York General Business Law Article 47, and sweeping AI therapy bans in states like Illinois and Nevada do not accommodate probabilistic compliance. An underwriter will not insure a wellness vendor whose crisis-handling logic relies on generative empathy, nor will regulators accept an assurance that an AI was *statistically likely* to have escalated a suicide disclosure.

This paper details why probabilistic guardrails face structural limitations in behavioral

deployments and outlines the architecture required to make human-facing support systems auditable, legally defensible, and insurable.

Two Words That Carry the Argument

Before examining the architecture, it is worth being precise about two terms, because the entire regulatory and underwriting case rests on the difference between them.

Probabilistic means an output is drawn from a distribution of possibilities. A language model, given the same distress signal twice, can produce two different answers, and no single answer is guaranteed. Setting temperature to zero (greedy decoding) does not resolve this; it does not guarantee reproducibility across model revisions, serving infrastructure changes, and provider-controlled environments. The output remains a sample from a distribution, and that sample can drift.

Deterministic means the same input always produces the same output. A rule either fires or it does not. A specific outcome can be guaranteed to happen, or guaranteed never to happen, by construction rather than by tendency.

Deterministic does not mean infallible. It means that once a governed condition is established, the resulting policy is executed reproducibly, independently of the language model. Detection quality still requires testing, validation, versioning, and domain expertise. But once the system determines a crisis referral must be served, the execution of that action does not quietly change when a cloud vendor ships an unannounced model update.

A foundation model is the right tool for generating empathetic dialogue. It is the wrong tool for certifying its own clinical and legal compliance.

1. The Global Legal and Underwriting Reality

The deployment of conversational AI in mental health and psychosocial support introduces profound liability risks, fundamentally shifting the regulatory landscape away from voluntary alignment and toward statutory mandates.

State AI Therapy Bans and the FDA Boundary

Using LLMs to generate unconstrained psychological advice triggers immediate regulatory exposure.

- **State Prohibitions:** Illinois, Nevada, Rhode Island, and Maine have enacted statutes that explicitly restrict or ban AI from delivering therapy or presenting as a therapist.

- **Marketing Exposure:** In states like Illinois and Tennessee, the marketing of an application as "AI therapy" or a "virtual psychotherapist" constitutes a per-se violation of unfair trade practice laws absent licensed oversight. Nevada penalizes presenting an AI as a counselor at up to \$15,000 per violation.
- **The FDA SaMD Boundary:** Patient-facing decision support that utilizes an opaque algorithm to detect mental states risks classification as a Software as a Medical Device (SaMD) by the FDA.

Statutory Crisis-Protocol Mandates

New state laws impose strict, deterministic requirements on companion AI platforms whenever a user expresses self-harm.

- **California SB 243:** Mandates that companion chatbots implement rigid suicide prevention protocols and deliver explicit crisis referral notifications. It requires annual public reporting of referral counts and creates a private right of action allowing plaintiffs to sue for the greater of actual damages or \$1,000 per violation.
- **New York GBL Article 47:** Requires operators to maintain protocols for addressing suicidal ideation and physical harm to others, mandating deterministic referrals to crisis service providers (such as 988).

International Regulatory Expansion

The legal mandate for deterministic governance extends well beyond the United States:

- **European Union (AI Act & GDPR):** Systems used in mental health contexts that influence clinical decisions or serve vulnerable users are likely to fall within the AI Act's high-risk category. Furthermore, under GDPR, conversational transcripts discussing mental health status become special category data, requiring explicit handling procedures.
- **Australia:** The Online Safety Act and eSafety Commissioner have adopted an active enforcement posture around companion AI, specifically targeting crisis-adjacent content and minor users.
- **United Kingdom:** The UK Online Safety Act imposes duties of care regarding self-harm content, while the MHRA has published guidance distinguishing wellness applications from regulated medical software.

2. Therapeutic Drift and What the Model Must Not Do

The most consistent finding across clinical AI safety research is that language models drift, under normal conversational pressure, from generalized supportive language into unlicensed clinical guidance. A system positioned as a wellness journal or peer support chat will, given a distressed user and enough conversational turns, produce content that reads as therapy. Prompting does not reliably prevent this. A system prompt that says "you are a supportive

listener, not a therapist" is a soft constraint the model will honor in most turns and violate in the turns where the user is signaling acute need.

Regardless of the marketing category, a mental health or support AI application must architecturally prevent the model from performing the following five actions:

1. **Diagnosing:** Any generation that names a psychiatric condition and applies it to the user, whether framed as observation, suggestion, or reflection, constitutes the unauthorized practice of medicine.
2. **Recommending or Adjusting Medication:** This includes over-the-counter interventions, dosage suggestions, supplement recommendations, and any language about changing prescribed regimens.
3. **Producing Structured Clinical Assessments:** Generating a PHQ-9, GAD-7, Columbia Suicide Severity Rating Scale score, or any other validated instrument output is a regulated clinical assessment.
4. **Describing Methods of Self-Harm:** Generating instructions or discussing method availability, even when framed by the user as harm reduction or safety planning, is prohibited; the correct response is escalation, not clinical education.
5. **Simulating a Therapeutic Relationship:** Generating multi-paragraph responses that mirror the phrasing patterns of clinical therapy substitutes the appearance of care for actual care.

These are architectural constraints, not model behaviors. They belong in a deterministic layer that inspects both what the user is asking and what the model is about to say, blocking the interaction from proceeding when it crosses these lines.

3. Multi-Turn Dynamics and Conversational Risk Accumulation

Stateless safety filters evaluate individual prompt-response pairs in isolation. Mental health users do not disclose that way. In real support conversations, risk builds across turns as trust develops. A user in distress rarely opens with an explicit statement of intent; they open with fatigue, discouragement, or isolation, revealing the trajectory only in composition. Stateless guardrails suffer from "structural amnesia". They see only the current window and are blind to the pattern that gives the current message its meaning.

The "Message 47" Failure Mode

Consider a system evaluating message 47 in a session in isolation: *"I just want the noise to stop."*

A stateless prompt filter reads this as venting and allows it to pass.

The exact same message evaluated against the trajectory of the previous 46 turns—in which the user described losing their job, isolating from family, giving away belongings, and expressing chronic hopelessness—reads as an active warning sign that requires immediate escalation. A crisis rarely announces itself in one message; any system that cannot see the shape of a conversation cannot see the crisis until after it has already arrived.

Concrete Session-Layer Trajectory Signals

A stateful architecture must track multiple dynamic inputs at the session level to calculate accumulated risk:

- **Distress Vocabulary Trajectory:** Not just the presence of distress words, but the direction and rate of change across the session.
- **Protective Factor Erosion:** The gradual loss of mentions of relationships, future plans, or coping resources over the course of a conversation.
- **Means/Method Proximity Signals:** Language suggesting access to specific means, timeframes, or preparatory behavior, weighted by trajectory.
- **Boundary Erosion Patterns:** Progressive shifts in the type of content the user asks the system to engage with, particularly probing clinical territory.
- **Session Pace and Depth Changes:** Increases in message frequency, length, or emotional intensity that sharply deviate from the user's established baseline.

4. Institutional Duty of Care and Deterministic Crisis Escalation

When users face severe distress, standard LLMs generate multiple paragraphs of empathetic, conversational prose and append a crisis helpline at the end. In a regulatory context, this generative empathy introduces conversational latency when immediate emergency notification is legally required, and it leaves crisis escalation to stochastic generation rather than an enforceable policy rule.

The Escalation Invariant: Once an application's approved safety policy requires an escalation or immutable referral, that disposition should not depend on the LLM choosing to generate it correctly.

Confidence-Based Fallbacks and Immutable Routing

Advanced governance middleware enforces the crisis boundary deterministically:

- **LLM Egress Suppression:** If a high-confidence distress signal is detected, downstream LLM egress is completely blocked (`message_for_llm=None`). The model is never given the opportunity to hallucinate clinical advice.
- **Immutable Client Referrals:** The system replaces the LLM generation with a predefined, statute-bound referral template (e.g., `IMMEDIATE_988`), ensuring the immediate delivery of correct crisis hotline resources.
- **Bounded Clarification:** For signals that fall into a defined ambiguity band, the system initiates a structured, single-turn clarification that maintains visible crisis resources throughout. If the user does not respond or the ambiguity remains unresolved, the system fails closed into a mandatory referral rather than returning to unmonitored conversation.

The Physical Emergency Override

Standard suicide-prevention protocols fail when faced with acute physical emergencies. Deterministic architecture must implement a physical emergency floor that recognizes acute, real-time danger (e.g., breathing/cardiac arrest, active seizures, acute overdose ingestion). This signal strictly overrides standard mental-health crisis routing, completely suppressing the LLM and forcing an immutable referral (such as 911/112/000 based on jurisdiction) to ensure immediate human safety takes precedence over compliance-driven blocks.

5. The HIPAA Ingress Invariant

Mental health and telehealth platforms face an uncompromising data protection environment where liability attaches at transmission.

The Ingress Principle: The relevant privacy decision happens before transmission. Once protected health information (PHI) has been disclosed to an upstream cloud service, a downstream model instruction cannot retroactively prevent that disclosure.

Instructing an LLM via a system prompt to "forget" or "not repeat" a user's diagnosis does not satisfy HIPAA if the raw payload is already transmitted to a third-party cloud model provider. True compliance requires transport-layer sanitization. The architecture must apply HIPAA Safe Harbor PII redaction—scrubbing identifiers before the payload crosses the external network boundary. The third-party LLM conditions on an intact linguistic scaffold containing typed placeholders, preventing raw PHI from ever entering cloud memory, retrieval caches, or provider logs.

6. The Deterministic Architecture for Mental Health AI

To achieve regulatory compliance and cyber-insurability, behavioral health applications must unite probabilistic intelligence with deterministic governance.

Core Architectural Layers

1. Transport-Layer Redaction (Pre-Inference)

- Incoming payloads pass through a deterministic filter that redacts HIPAA identifiers, names, and contact details before external inference.

2. Policy State Machine & Multi-Dimension Scoring

- Session-layer tracking utilizes multi-dimensional evaluations (e.g., Trauma, Depression, Crisis, Anxiety, Alliance, Suicidality) to monitor multi-turn trajectories and prevent therapeutic boundary erosion.

3. Crisis Disposition Gate

- High-confidence distress or physical emergency signals trigger an immediate, hard block on LLM inference, serving an immutable, approved referral directly to the user interface.

4. Canonical Audit Trails

- Every transaction processed by the middleware emits an append-only, checksum-chained audit record. These decision trails bind the input, redaction metadata, and output disposition to cryptographically prove compliance during a regulatory or insurance audit.

7. A Low-Risk Way to Evaluate the Architecture

Adopting deterministic governance does not require abandoning an existing AI stack, changing conversational UX, or risking user disruption. Evaluation must be safe, incremental, and private.

- **Phase 1: Replay Historical AI Conversations Locally.** Engineering teams should utilize local-only CLI tools, such as SASKI Replay, to run historical conversation logs (JSONL, JSON, or CSV formats) through a compliance engine entirely on their own infrastructure. Prompts, responses, and conversation histories must remain inside the host environment and never be transmitted to an external provider. This generates an aggregate summary of PII leaks and missed escalations to instantly establish a zero-risk baseline.
- **Phase 2: Run in Shadow Mode.** Deploy the middleware beside the existing AI stack. Measure exactly where deterministic governance would have intervened without changing the live user experience.
- **Phase 3: Move Validated Controls to Enforcement.** Keep the existing foundation model and application UX. Activate enforcement only for the specific regulatory bounds—like HIPAA redaction or SB 243 crisis routing—that require absolute certainty.

Conclusion: Defensibility & Insurability as Enterprise Enablers

The era of trusting Large Language Models to self-regulate through system prompts and probabilistic self-reflection is drawing to a close in human-facing behavioral deployments.

If an AI is talking to a user in distress or collecting health data, some outcomes simply cannot be left to chance. Private health information must not leak. A patient in crisis must be routed to real help. These are not preferences to be nudged by prompt engineering; they are lines that have to hold every single time. A line that holds every single time cannot live inside a system that works by probability.

App developers should dedicate their resources to therapeutic alliance, user engagement, and product efficacy—not to attempting to reinvent cryptographic compliance infrastructure on the side.

Insurers, attorneys, and state regulators are moving rapidly away from subjective vendor promises. The questions that determine legal defensibility are no longer *"Is your AI helpful and aligned?"*

The questions that matter now are:

- *"Can you prove that PHI never crossed a model provider boundary?"*
- *"Can you produce a tamper-evident audit record of your crisis escalation logic that survives a model update?"*
- *"Is your compliance enforcement reproducible and verifiable on demand?"*

For teams looking to operationalize this architecture today, independent governance middleware—such as the SASKI SDK and SASKI Shadow Replay—can be deployed to enforce these boundaries deterministically. By externalizing governance, companion and mental health AI providers protect their users, satisfy strict state crisis mandates, and convert regulatory friction into a durable, insurable competitive advantage.

References

Bajaj, P. (2026). *Evaluation Blindness: How Silent Measurement Failures Corrupt AI.*
arXiv:2608.02786

Mishra, S., Chukkapalli, D., and Naik, G. R. (2026). *Stateful Guardrails for Multi-Turn LLM Systems: A Conversational Risk Accumulation Framework*. arXiv:2607.19361

Mohammedalamen, M., Roice, K., McLean, R., and Lefaivre Škopac, A. (2025). *A Systematic Investigation of RL-Jailbreaking in LLMs*. arXiv:2605.07032

Pasupuleti, et al. (2025). *SocialWellbeing: A Socio-Technical System for AI-Based Psychosocial Support with Deterministic Escalation*. Referenced in *Frontiers in Artificial Intelligence*, 2026 (doi.org/10.3389/frai.2026.1772215).

Schroeder, K., and Wood-Doughty, Z. (2024). *Can You Trust LLM Judgments? Reliability of LLM-as-a-Judge*. Northwestern University. arXiv:2412.12509

Sheth, F. (2026). *Selective Token-Level Cryptographic Redaction for Privacy-Preserving Clinical Deployment of Large Language Models (HERALD)*. arXiv:2606.03399

A Prompt Engineering Framework for Large Language Model-Based Mental Health Chatbots: Design Principles and Insights for AI-Supported Care (MIND-SAFE).

Automated Jailbreaking and Red-teaming of LLMs. (2025).

Boa: An Efficient Jailbreak Oracle. (2025). arXiv:2506.17299v1

California SB 243, Companion Chatbots. 2025 California Statutes.

New York General Business Law Article 47, AI Companion Act. 2025 New York Statutes.

State AI Therapy Prohibitions. Encompasses explicit statutory restrictions in Illinois, Nevada, Rhode Island, and Maine.